
EDUCATION

University of Massachusetts Amherst | College of Information and Computer Sciences

Bachelor of Science in Computer Science, May 2025

Coursework: Programming with Data Structures, Introduction to Algorithms, Operating Systems, Computer Networks, Software Engineering, Machine Learning, Artificial Intelligence, Elementary Biochemistry, Introductory Biology I and II, Organic Chemistry I and II

Honors: Dean's List honors Fall 2021, Dean's List honors Spring 2022, Dean's List honors Spring 2023

PROJECTS

Operations Tracker - *FastAPI, Python, SQLAlchemy, PostgreSQL, Docker*

- Built a RESTful backend for tracking projects with full CRUD, input validation, and unique name enforcement; exposed auto-generated OpenAPI docs.
- Containerized PostgreSQL with Docker Compose and environment-based config, implemented dependency injected DB sessions and reliable connection handling.

Custom ML Benchmarking Suite - *Python, NumPy, Matplotlib, scikit-learn*

- Built a full benchmarking framework evaluating k-NN, Random Forests, and Neural Networks across four datasets with mixed numerical and categorical features, implementing custom normalization and k-fold cross-validation.
- Achieved strong results across models, including 98.55% accuracy (KNN), 93.05% accuracy (Random Forest), and 92.8% accuracy (Neural Network) on datasets.

Gesture-Controlled Drone System - *Python, SVM, ESP32, IMU, Tello SDK*

- Built a machine learning pipeline for drone gesture control by collecting 420 IMU sequences (7 gestures, 3 users), engineering FFT-based features, and training an RBF SVM that achieved 94% multi-class accuracy.
- Developed a real-time system that streamed IMU data from an ESP32, performed low-latency inference, and triggered Tello drone maneuvers including movement, rotation, and flips.

Spotify Playlist Generator - *Node.js, JavaScript, Express, Spotify API, React*

- Built and maintained a full-stack playlist generator using OAuth Authorization Code Flow, session-based authentication, and 5 backend REST endpoints for search, audio-feature recommendations, and playlist creation.
- Designed a recommendation engine by aggregating Spotify audio features from up to 5 user-selected tracks, producing personalized track suggestions via the Spotify Web API.
- Refactored a React frontend into 8 reusable components, including a drag-and-drop interface for selecting tracks and generating playlists directly in the user's Spotify account.
- Improved reliability and maintainability by refactoring backend routes and frontend state management, adding clearer documentation and stronger error handling for Spotify API requests.

WORK EXPERIENCE

Lowe's - *Customer Service*

July - August 2023

- Provided customer support in a fast-paced retail environment, addressing needs and restocking inventory.

Busy Bee Jumpers - *Driver Assistant*

June - August 2022, May - August 2024

- Provided setup and teardown of event equipment, including tents, inflatables, tables, and chairs.

SKILLS

Languages: Java, Python, TypeScript, JavaScript, C, C++, HTML/CSS

Frameworks & Tools: React, Node.js, FastAPI, SQLAlchemy, Docker, Git, GitHub

Data & ML: NumPy, Pandas, scikit-learn, Matplotlib, SQL (PostgreSQL/MySQL)